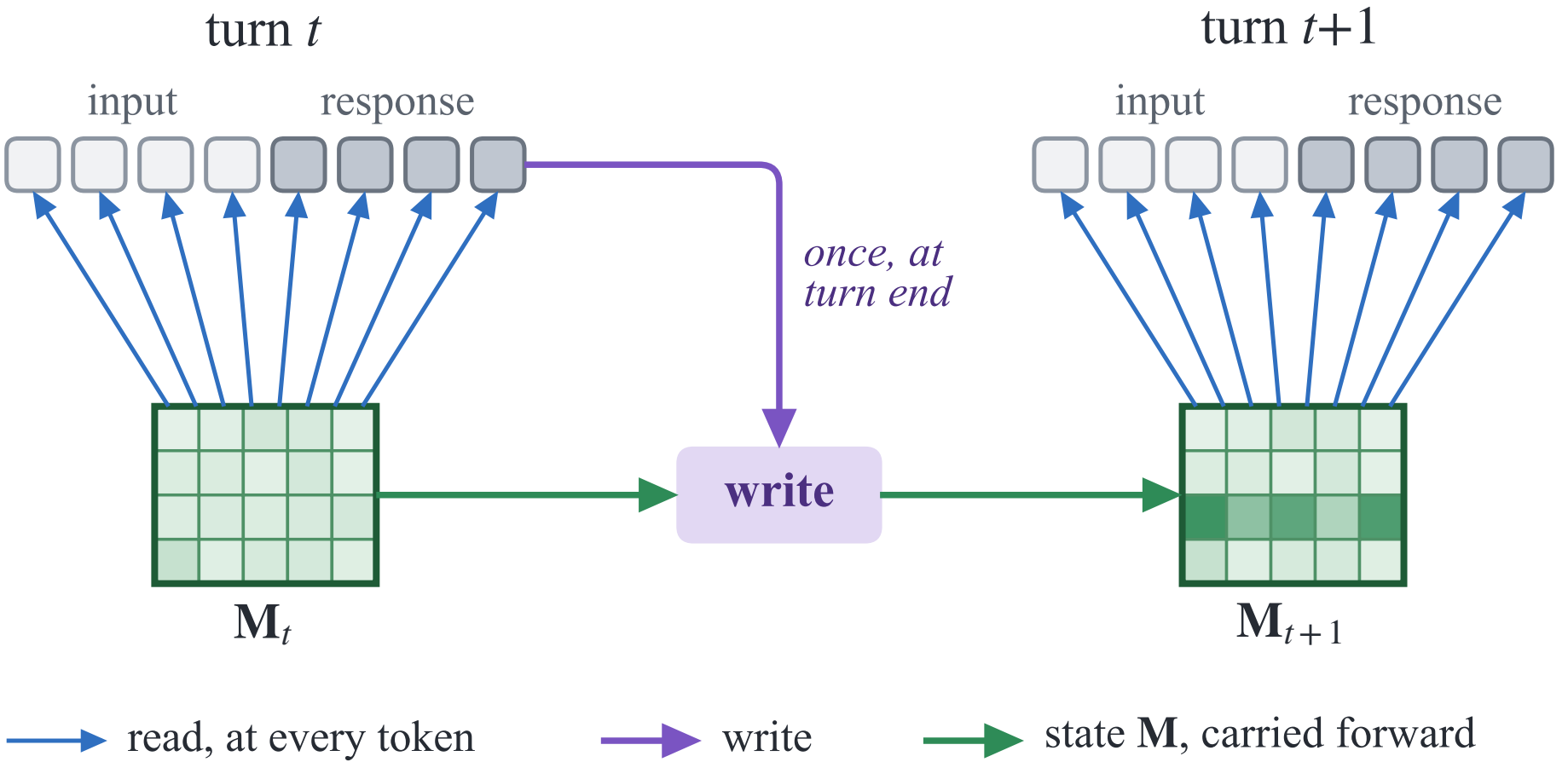
turn t
input
response
turn t+1
input
response
once, at
turn end
write
$\mathbf{M}_t$
$\mathbf{M}_{t+1}$
read, at every token
write
state **M**, carried forward